

DEPARTMENT OF COMPUTER SCIENCE

Data Structures & Algorithms - Semester IV

Time: 3 hours | Maximum marks: 60

Answer all 12 questions. Each question carries 5 marks.

- Q1. Define an array and state its memory representation. [5 marks] [Unit 1] [CO1]
- Q2. Explain the working of breadth first search. [5 marks] [Unit 4] [CO4]
- Q3. Apply binary search to find 23 in the sorted array [4, 9, 15, 23, 31]. [5 marks] [Unit 1] [CO1]
- Q4. Analyze the time complexity of quicksort in the worst case. [5 marks] [Unit 5] [CO5]
- Q5. List the basic operations on a stack. [5 marks] [Unit 2] [CO2]
- Q6. Explain the differences between a stack and a queue. [5 marks] [Unit 2] [CO2]
- Q7. Compare a binary search tree and an AVL tree. [5 marks] [Unit 3] [CO3]
- Q9. Evaluate the suitability of an adjacency list for a sparse graph. [5 marks] [Unit 4] [CO4]
- Q10. Design a hash table with collision handling for a student registry. [5 marks] [Unit 5] [CO5]
- Q11. Explain the working of breadth first search. [5 marks] [Unit 4] [CO4]
- Q12. Analyze how tree balance affects search performance. [5 marks] [Unit 3] [CO3]
- Q13. Describe the process of inserting an element into a linked list. [5 marks] [Unit 1] [CO1]